DS	**Session**	Sorting	**Question Information**	Level 1 Challenge 12

Question description

Sajid is an third year student in a reputed institution.

Although he scored well in many subjects, he did not an expert in computer programming languages.

But Sajid's computer examination is scheduled for next week.

As per the blueprint, many questions would come from the sorting topic.

He collected previous year's questions. one of the repeated questions is to sort the given set of numbers using Selection Sort

The first line of the input contains the number of elements N, the second line of the input contains the numbers A_i to be sorted.

In the output print the the final sorted array in the given format.

Can you help him ?

Constraints

$1 <= N <= 10^5$
$1 <= A_i <= 10^9$

Input:
The first line of the input contains the number of elements

the second line of the input contains the numbers to be sorted.

Output:
print the the final sorted array in the given format.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
15 17 11 25 1
```

EXPECTED OUTPUT

```
1 11 15 17 25
```

Test Case 2

INPUT (STDIN)

```
6
65 45 27 21 25 11
```

EXPECTED OUTPUT

```
11 21 25 27 45 65
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void selectionSort(int arr[],int n)
```

Test Case 2

KEYWORD

```
void swap(int *xp,int *yp)
```

Test Case 3

KEYWORD

```
void printArray(int arr[],int size)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

47

Code

solution.c

```c
#include <stdio.h>

void swap(int *xp, int *yp) {
    int temp = *xp;
    *xp = *yp;
    *yp = temp;
}

void selectionSort(int arr[], int n) {
    int i, j, min_idx;
    for (i = 0; i < n - 1; i++) {
        min_idx = i;
        for (j = i + 1; j < n; j++) {
            if (arr[j] < arr[min_idx]) min_idx = j;
        }
        swap(&arr[min_idx], &arr[i]);
    }
}

void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        printf("%d%s", arr[i], (i == size - 1) ? "" : " ");
    }
    printf("\n");
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    selectionSort(arr, n);
    printArray(arr, n);
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell

PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
6
65 45 27 21 25 11
11 21 25 27 45 65
PS C:\C Programming>
```